

Observations

(multi-view images)

Task: Is there a TV in the room?



RGB-D Images



Memory Representation

(vantage-aware)



Position: Bedroom
Evidence: Partially detected



Position: Living room
Evidence: Fully detected



Position: Kitchen
Evidence: Partially detected

Supporting Views



Structured Memory



Claim: TV exists.



Support Set: v1, v2, v3



Support Score: 3



Estimate Position: (x, y, z)

Each claim retains supporting views and source poses.

Selective Decision

(evidence-sufficient)

No Support v0

TV is not visible

No

Adapter

Low Support v1 v3

TV is partially visible

? Abstain

High Support v2

The entire TV is visible

Cross-view spatial consistency

High score

Yes

Wait for more coverage and additional views